

Fractal Model of Elastic-Plastic Contact Between Rough Surfaces

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Roughness measurements by optical interferometry and scanning tunneling microscopy on a magnetic thin-film rigid disk surface have shown that its surface is fractal in nature. This leads to a scale-dependence of statistical parameters such as r.m.s height, slope and curvature, which are extensively used in classical models of contact between rough surfaces. Based on the scale-independent fractal roughness parameters, a new model of contact between isotropic rough surfaces is developed. The model predicts that all contact spots of area smaller than a critical area are in plastic contact. When the load is increased, these plastically deformed spots join to form elastic spots. Using a power-law relation for the fractal size-distribution of contact spots, the model shows that for elastic deformation, the load P and the real area of contact A_r are related as $P \sim A_r^{(3-D)/2}$, where D is the fractal dimension of a surface profile which lies between 1 and 2. This result explains the origins of the area exponent which has been the focus of a number of experimental and theoretical studies. For plastic loading, the load and area are linearly related. The size-distribution of spots also suggests that the number of contact spots contributing to a certain fraction of the real area of contact remains independent of load although the spot sizes increase with load. The model shows that the load-area relation and the fraction of the real area of contact in elastic and plastic deformation are quite sensitive to the fractal roughness parameters.

1 Introduction

The contact between two rough surfaces has strong influence on the phenomena of friction, stiction, wear, and lubrication as well as conduction of heat and electricity. The presence of surface roughness results in an imperfect contact which leads to the real area of contact being a fraction of the apparent or nominal area. Early studies of this phenomenon (Archard, 1957; Bowden and Tabor, 1964) have shown that the real area of contact A_r and the compressive load P on the surface follow the power-law relation $P \sim A_r^n$. The origins and the value of the exponent n were the major focus of most studies and are as yet not completely understood. The Amontons's law of friction suggests that n is close to unity. This could either be due to purely plastic deformation or due to certain geometric arrangements of asperities in elastic deformation (Archard, 1957). This paper adopts a fundamental approach to develop a new model of contact and attempts to explain the physical meaning of the exponent.

Roughness measurements of an untextured thin-film magnetic rigid disk of type C (Bhushan and Doerner, 1989) by an optical interferometer (Majumdar and Bhushan, 1990) of resolution $1\mu\text{m}$ and by a scanning tunneling microscope of resolution 1nm , are shown in Fig. 1. The figure shows that roughness is found at both millimeter and nanometer scales. Smooth surfaces are rare in nature and most surfaces are found to be rough over several length scales ranging from macroscales, such as that of the earth, to nanoscales (Avnir et

al., 1984). It can be concluded that a surface is composed of a large number of length scales of roughness that are superimposed on each other.

Conventionally, the deviation of a surface $z(x,y)$ from its mean plane $z(x,y)=0$ is assumed to be a random process and characterized by statistical parameters such as the standard deviations of the surface height σ , slope σ' and curvature σ'' (Nayak, 1971). The theories of elastic (Greenwood and Williamson, 1966; McCool, 1986) and plastic (Nayak, 1973; Gupta and Cook, 1972; Bhushan, 1990) contact between rough surfaces use these parameters for their analysis. However, due to the multiscale nature of surfaces it is found that the variances of slope and curvature depend strongly on the resolution of the roughness-measuring instrument or any other form of filter, and hence not unique for a surface (Bhushan et al., 1988). It is also known that surface topography is a nonstationary random process for which the variance of the height distribution depends on the length of the statistical sample (Sayles and Thomas, 1978). The dependence of all three parameters on the spatial resolution and magnification of a magnetic tape surface (Majumdar and Bhushan, 1990) is shown in Fig. 2. This plot also shows the variation of r.m.s values of peak heights, peak slopes and peak curvatures with the magnification. These parameters are also often used in modeling contact between two surfaces (Bhushan, 1984a). The scale-dependence in Fig. 2 suggests that instruments with different resolutions and scan lengths yield different values of these statistical parameters for the same surface. Therefore the predictions of the contact models

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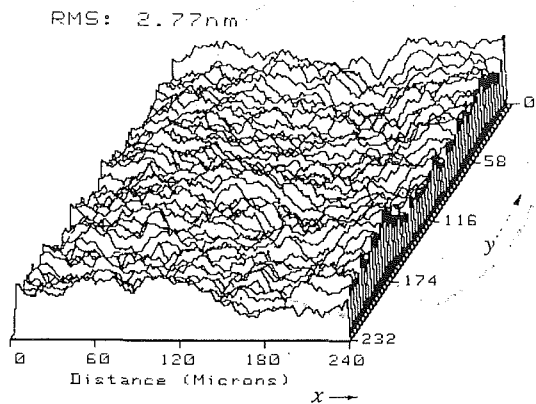


Fig. 1(a)

based on these parameters may not be unique to a pair of rough surfaces.

By using r.m.s height, slope and curvature, previous contact theories consider only an *averaged* surface with a single scale of roughness to be in contact with a smooth plane. Therefore, they fail to consider the multiscale nature of rough surfaces in the contact process. However if a rough surface is characterized in a way such that the structural information of roughness of all scales is retained, then it will be more logical to use such a characterization in a contact theory. In order to develop a contact theory based on this motivation, it is first necessary to quantify the multiscale nature of surface roughness.

A unique property of rough surfaces is that if a surface is repeatedly magnified, increasing details of roughness are observed right down to nanoscales as observed in Fig. 1. In addition, the roughness at all magnifications appear quite similar in structure as qualitatively shown in Fig. 3. In recent papers (Majumdar and Bhushan, 1990; Majumdar and Tien, 1990), it was established that such a behavior can be characterized by *fractal geometry*. Ling (1987, 1989) has also established that the total length of a contoured surface profile depends on the scale of measurement and follows a scaling relation that suggests a fractal behavior. The main conclusions from these studies were that a fractal characterization of surface roughness is *scale-independent* and provides information of the roughness structure at all the length scales that exhibit the

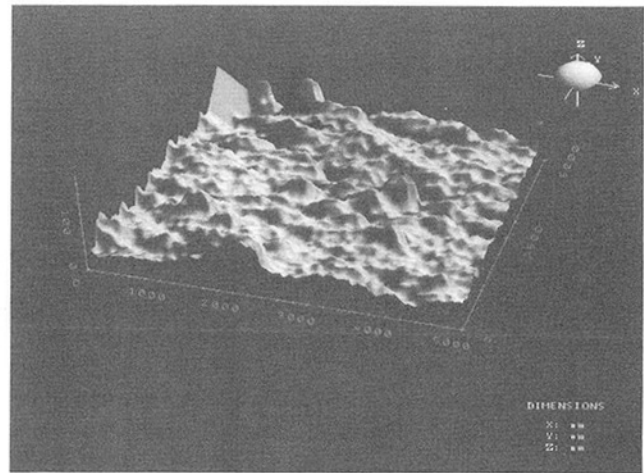


Fig. 1(b)

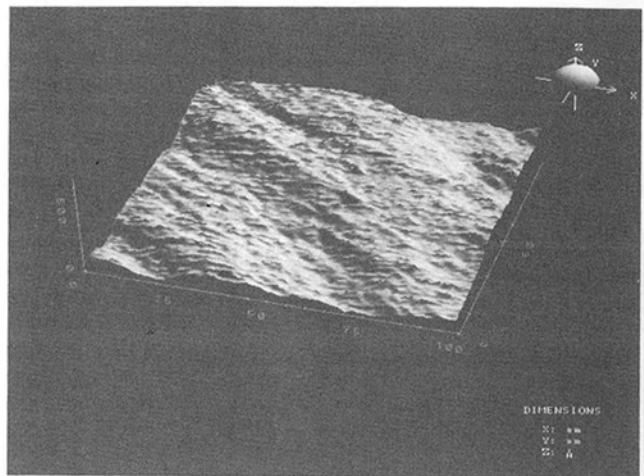


Fig. 1(c)

Fig. 1 Surface topography of a magnetic thin-film rigid disk of type C (Bhushan and Deorner, 1989) at different length scales (a) $240\mu\text{m} \times 240\mu\text{m}$ measured by an optical profiler (b) $5\mu\text{m} \times 5\mu\text{m}$ measured by an STM (c) $0.1\mu\text{m} \times 0.1\mu\text{m}$ measured by an STM

Nomenclature

a = area of a contact spot [m^2]
 A = area [m^2]
 C = scaling factor for spectrum, $S(\omega) = C^2 \omega^{-3}$ for $D=1$
 d = separation between surfaces [m]
 D = fractal dimension of a surface profile
 D_s = fractal dimensional of a surface, $D+1$
 E = elastic modulus [Nm^{-2}]
 f = fraction of the real area of contact
 G = fractal roughness parameter [m]
 K = factor relating hardness to yield strength
 l = length scale of an asperity or contact spot [m]
 L = characteristic length of a surface [m]

$n(a)$ = size-distribution of contact spots of area a
 N = number
 P = load [N]
 R = radius of asperity curvature [m]
 S = power of a profile spectrum [m^3]
 z = surface height [m]
 α = roughness parameter defined in equation (4)
 δ = amplitude or deformation of an asperity [m]
 ϕ = material property, σ_y/E
 γ = scaling parameter for the Weierstrass-Mandelbrot function
 ω = frequency or reciprocal of length scale [m^{-1}]
 σ = standard deviation of surface height [m]

σ' = standard deviation of surface slope
 σ'' = standard deviation of surface curvature [m^{-1}]
 σ_y = yield strength of softer surface [Nm^{-2}]

Subscripts

a = apparent or nominal
 c = critical
 e = elastic
 f = fraction
 h = higher cut-off
 l = lower cut-off or largest spot
 p = plastic
 r = real
 s = smallest spot

Superscripts

$*$ = nondimensional forms defined in equation (28)

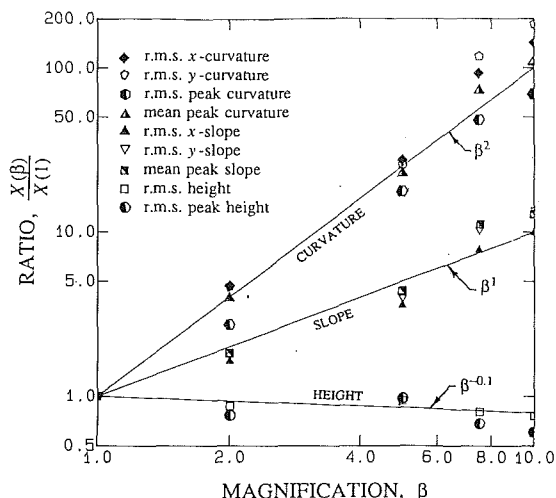


Fig. 2 Scale dependence of statistical roughness parameters for a magnetic tape (Majumdar and Bhushan, 1990). Magnification of 1 corresponds to a lateral resolution of $4\mu\text{m}$ and that of 10 corresponds to $0.4\mu\text{m}$.

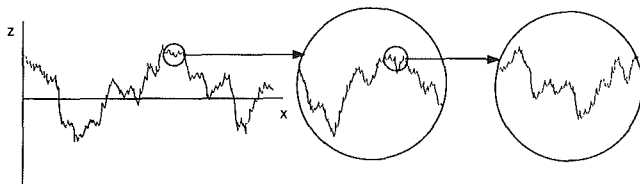


Fig. 3 Qualitative description of statistical self-affinity for a surface profile

fractal behavior. Based on this observation, a new theory of contact between rough surfaces is developed here.

This paper first briefly reviews the fractal characterization of surface roughness and presents experimental evidence of the fractal nature at micro- and nanoscales. A new model of contact is then developed. The results of this model are compared with previous experiments. The paper concludes with a critique of the fractal contact model emphasizing its merits and limitations.

2 Fractal Characterization of Surface Roughness

Consider a surface profile $z(x)$ as shown in Fig. 3 which appears random, multiscale, and disordered. The mathematical properties of such a profile are that it is continuous, nondifferentiable and statistically self-affine (Majumdar and Bhushan, 1990; Majumdar and Tien, 1990; Majumdar, 1989). The nondifferentiability arises from the fact that a tangent or a tangent plane cannot be drawn at any point on the surface since more and more details of roughness will appear at that point. In short, a rough surface is never smooth at any length scale. The statistical self-affinity is due to similarity in appearance of a profile under different magnifications. It was found that the Weierstrass-Mandelbrot W-M function satisfies all these properties and is given as (Majumdar and Bhushan, 1990; Berry and Lewis, 1980)

$$z(x) = G^{(D-1)} \sum_{n=1}^{\infty} \frac{\cos 2\pi\gamma^n x}{\gamma^{(2-D)n}}; \quad 1 < D < 2; \quad \gamma > 1 \quad (1)$$

where the parameter D is the fractal dimension, G is a characteristic length scale of the surface and γ^n determines the frequency spectrum of the surface roughness. The theory of fractal geometry and the concept of fractional dimension is well described by Mandelbrot (1982) and in the recent paper of Majumdar and Bhushan (1990). The parameters which

characterize the W-M function are G , D , and n_1 where $\gamma = 1.5$ was found to be a suitable value for high spectral density and for phase randomization (Majumdar and Tien, 1990). Since a rough surface is a nonstationary random process (Sayles and Thomas, 1978) the lowest frequency is related to the length L of the sample as $\gamma^{n_1} = 1/L$. The parameters G and D can be found from the power spectrum of the W-M function

$$S(\omega) = \frac{G^{2(D-1)}}{2 \ln \gamma} \frac{1}{\omega^{(5-2D)}} \quad (2)$$

where $S(\omega)$ is the power of the spectrum and ω is the frequency which is the reciprocal of the wavelength of roughness. Note that if $S(\omega)$ is plotted as a function of ω on a log-log plot, then the power law behavior would result in a straight line. The slope of the line is related to the fractal dimension D of the surface roughness and the parameter G is related to the location of the spectrum along the power axis.

It is important to note that in contrast with the classical parameters of r.m.s height, slope, and curvature, the characterization parameters G and D are independent of ω and therefore scale-independent. The relations between the classical and the fractal parameters are

$$\sigma = \left[\int_{\omega_l}^{\omega_h} S(\omega) d\omega \right]^{1/2} \quad (3a)$$

$$\sigma' = \left[\int_{\omega_l}^{\omega_h} \omega^2 S(\omega) d\omega \right]^{1/2} \quad (3b)$$

$$\sigma'' = \left[\int_{\omega_l}^{\omega_h} \omega^4 S(\omega) d\omega \right]^{1/2} \quad (3c)$$

While measuring surface topography by any technique, the lowest frequency ω_l is related to the length of the sample and the highest frequency ω_h depends on the resolution of the measuring instrument or any other form of filter. If it is assumed that $\omega_h \gg \omega_l$ then using equation (2) for the power spectrum it can be showed that $\sigma \approx \omega_l^{(D-2)}$, $\sigma' \approx \omega_h^{(D-1)}$ and $\sigma'' \approx \omega_h^D$ (Majumdar and Bhushan, 1990). Such a behavior can be observed in Fig. 2 for a value of $D \sim 2$. Thus it can be concluded that no matter how the roughness of a surface is measured the conventional statistical parameters will depend on the sample length and instrument resolution.

To verify whether surfaces do follow a power-law fractal behavior and to obtain the parameters D and G of a surface, one needs to compare the power spectrum of a real surface profile with that of the W-M function as given in equation (2). Figures 4(a) and 4(b) show the averaged spectrum of a surface profile of an untextured thin-film magnetic rigid disk of type C. The spectra in Fig. 4(a) corresponds to the surface shown in Fig. 1(a) that was measured by an optical profiler. The spectrum in Fig. 4(b) corresponds to the surface of Fig. 1(b) which was measured by a scanning tunneling microscope. The spectra in Fig. 4(a) follows $S(\omega) \sim \omega^{-2.24}$ corresponding to $D = 1.38$ whereas the spectrum in Fig. 4(b) follows $S(\omega) \sim \omega^{-2.35}$ corresponding to $D = 1.33$. The experiments show that the surface of the rigid disk follows a fractal structure for three decades of length scales.

It is important to note that Ling (1987, 1989) found an interesting scaling relation for roughness profiles. If $\mathcal{R}$ is the contour length of a surface profile normalized by the straight-line distance between two points, a scaling law of the form $\mathcal{R} = \exp(2.3(\epsilon/\epsilon_c)^{-\alpha})$ was observed for a variety of surfaces, where ϵ is the unit of measurement and α and ϵ_c are specific to a surface. The value of $\alpha = 2$ was found for a number of surface profiles. Although such a scaling law and the universal nature of α suggest a fractal-like behavior and may have deep-rooted physical implications, it is difficult to determine the fractal dimension from such a scaling law. This difficulty arises due to a mathematical reason of surface pro-

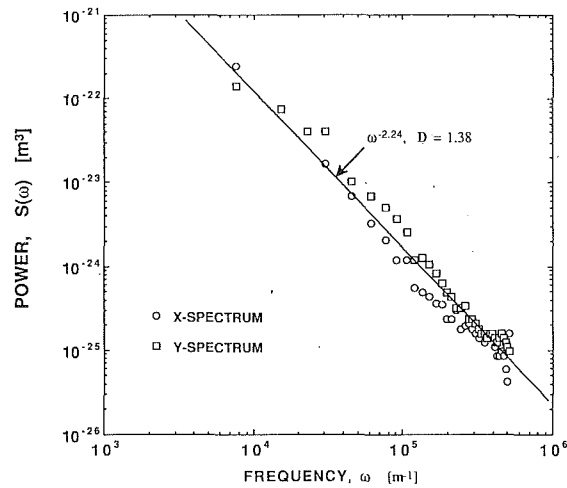


Fig. 4(a)

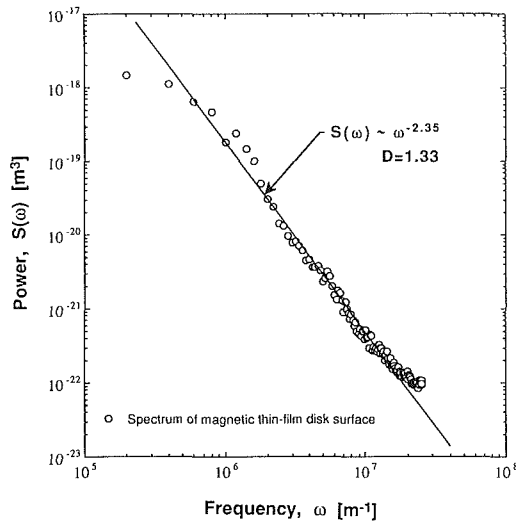


Fig. 4(b)

Fig. 4 Averaged power spectra of a magnetic thin-film rigid disk surface at different length scales corresponding to (a) Fig. 1(a) (b) Fig. 1(b)

files being *self-affine* and not *self-similar*. Mandelbrot (1967) determined the length of the coastline of Britain for different units of measurement and concluded that the coastline is a statistically self-similar fractal. This implied that for *equal* magnification of the coastline in all directions, the coastline images appeared similar. In contrast, the spectra of Figs. 4(a) and 4(b) suggests that surface profiles are *self-affine*, which means that the scaling in the vertical and lateral directions of a profile are different. This is evident in the scaling relation $S(\gamma\omega) \sim \gamma^{(5-2D)} S(\omega)$ which shows that when the spatial frequency is scaled by γ , the power of the spectrum scales as $\gamma^{(5-2D)}$. It has been pointed out by Mandelbrot (1985) and Voss (1988) that for self-affine profiles, the contoured length of the profile does not provide information of the fractal dimension. For self-affine profiles, the quantity of interest is the structure function $\mathcal{V}(\tau) = \langle (z(x+\tau) - z(x))^2 \rangle$, which is related to the power spectrum by a Fourier transform (Majumdar and Bhushan, 1990).

The basic idea behind a fractal characterization is that instead of characterizing the actual disorder of surface roughness as the classical statistical methods do, it is more logical to identify and characterize the order behind such a disorder. The order in this case is the power-law behavior of the spectrum which on a log-log plot yields a straight line. The power law behavior of the spectrum suggests a scaling relation of the form

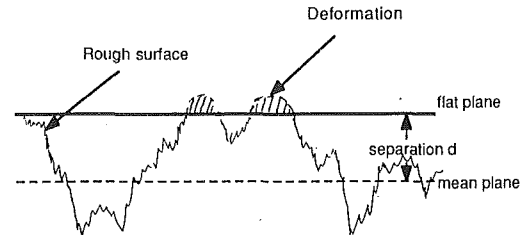


Fig. 5(a)

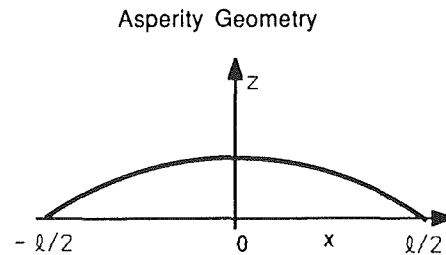


Fig. 5(b)

Fig. 5 (a) Contact between a rough surface and a flat plane producing isolated contact spots and (b) Geometry of a contact spot of length scale l

$$\delta = G^{D-1} l^{2-D} \quad (4)$$

between the roughness amplitude δ and the length scale l , which can be obtained from the W-M function for a single frequency mode of $\gamma^n = 1/l$. The amplitude δ in this case is with respect to the local mean over a sample length l . Theoretical studies on surface formation and growth (Kardar et al., 1986) also suggest this relation for roughness amplitude where the value of the fractal dimension is universal for a growth process and is related to the physics of surface formation. Note that once the parameters G and D are known, the roughness amplitude can be obtained for any length scale within the fractal regime. This is particularly important in studying the effects of surface roughness on any physical phenomenon at the relevant length scales.

3 Fractal Contact Model

The fractal model developed here is based on the spectrum of a surface profile which has been experimentally seen to follow the power-law behavior of equation (2). The parameters which characterize the spectrum are G , D and ω_l , where ω_l depends on the characteristic length of the surface L , $\omega_l = 1/L$. The length L is related to the apparent or nominal area A_a by the relation $A_a = L^2$. It is assumed that the roughness is statistically isotropic and that during contact, the interactions between neighboring asperities, work hardening due to plastic-elastic transition, variation of material hardness with depth from the surface and the frictional forces between deforming asperities are negligible. The Greenwood-Williamson model, in addition, assumes that all asperities have equal radii of curvature. It is shown here that the radius of curvature depends on the size of contact and is incorporated in the fractal model. In the following discussion it will be clear that the dependence of hardness on the depth can be incorporated in the model developed now.

Elastic-Plastic Regime of Contact Spots. Consider the interface between a statistically isotropic rough surface and a flat plane as shown in Fig. 5(a). The contact spots will be of different sizes and spread randomly over the contact interface. Let the size-distribution of contact spots be $n(a)$, such that $n(a)da$ is equal to the number of contact spots of area between a and $a+da$. The real area of contact is related to $n(a)$ as follows

$$A_r = \int_{a_s}^{a_l} n(a) da \quad (5)$$

where a_l and a_s are the area of the largest and smallest spots, respectively. Now consider one spot of area a and the characteristic length[†] scale $l = a^{1/2}$. The idea of the fractal model is to first determine the contact spot area a , then from equation (4) predict the deformation required to obtain that spot area and finally predict the load required for such a deformation. For a given contact spot of area a , the height or deformation is $\delta = G^{D-1} a^{(2-D)/2}$. The spectrum suggests that the roughness is composed of sinusoidal waves of different wavelengths and amplitudes superimposed on each other. From the W-M function, it is clear that for a wavelength of $l = \gamma^n$, the roughness is a cosinusoidal wave for which the asperity shape before deformation is given as

$$z(x) = G^{D-1} l^{(2-D)} \cos\left(\frac{\pi x}{l}\right); \quad -\frac{l}{2} < x < \frac{l}{2} \quad (6)$$

as shown in Fig. 5(b). The radius of curvature R , at the tip of the asperity is given by

$$R = \left| \frac{1}{\frac{d^2 z}{dx^2}} \right|_{x=0} = \frac{l^D}{(\pi)^2 G^{D-1}} = \frac{a^{D/2}}{(\pi)^2 G^{D-1}} \quad (7)$$

Depending on the radius of curvature R and the deformation δ , the contact spot will be either in elastic or plastic deformation. The critical deformation δ_c , for the inception of plastic contact of a hemisphere has been shown to be (Greenwood and Williamson, 1966)

$$\delta_c = \left(\frac{\pi K \sigma_y}{2E} \right)^2 R \quad (8)$$

where σ_y is the yield strength of the softer material, E is the composite elastic modulus from Hertzian analysis (Johnson, 1985) and K is a factor relating the hardness H to the yield strength σ_y , as $H = K\sigma_y$. Bhushan (1984b) lists the different values of K ranging from 0.5 to 2 that previous investigations have adopted. Using equation (7) for R it follows that

$$\delta_c = \left(\frac{K\phi}{2} \right)^2 \frac{a^{D/2}}{G^{D-1}} \quad (9)$$

where $\phi = \sigma_y/E$ is a material property. If $\delta > \delta_c$, then the asperity is in plastic contact. Using equation (4) for δ and equation (9) for δ_c , the range of contact spots under plastic contact can be determined. It is interesting to note that when the expressions for δ and δ_c are used in the inequality, it is found that for a contact spot of area a to be in plastic deformation, it should satisfy the relation $a < a_c$ where a_c is the critical area demarcating the elastic and plastic regimes and is given by

$$a_c = \frac{G^2}{(K\phi/2)^{2/(D-1)}} \quad (10)$$

This shows that small contact spots are in plastic contact whereas large spots are in elastic contact. This result is in contrast with that of the G-W model (Greenwood and Williamson, 1966) where larger spots are found to be in plastic contact and small ones are in elastic deformation. The reason for such a counter-intuitive result is that although the deformations of a small contact spot is small, its radius of curvature depends on the area of contact spots by equation (7) and is therefore

[†] It is assumed that the contact spot is a circle and therefore there exists only one characteristic length. This is reasonable when the roughness is statistically isotropic. For anisotropic surfaces, the contact spot can be modeled as an ellipse for which there are two characteristic lengths—the major and minor axes.

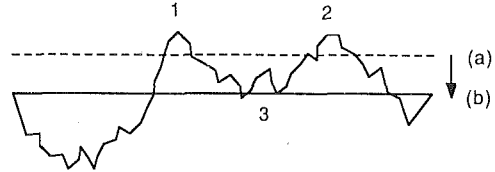


Fig. 6 Graphical depiction of the contact process when contact plane moves from position (a) to position (b)

also small. The smaller radius of curvature results in a lower value of the critical deformation δ_c . It is important to note that since both R and δ_c vary as $a^{D/2}$ and δ varies as $a^{(2-D)/2}$, the condition $\delta > \delta_c$ is satisfied only by smaller contact spots of area $a < a_c$.

Since a_c plays an important role in deciding the range of plastic and elastic deformation, it is worth investigating the effects of D , G and ϕ on a_c . Note that a_c is obtained by equating δ and δ_c as derived in equation (10). Consider the limiting case of $D=1$ for which the spectrum is $S(\omega) = C^2/\omega^3$. For such a case $\delta = Ca^{1/2}$ and $\delta_c = (K\phi/2)^2 a^{1/2}/C$. Since both δ and δ_c vary as $a^{1/2}$, it is interesting to note that if $C > K\phi/2$, then all contact spots are plastic and if $C < K\phi/2$ then all spots are in elastic deformation. For the limiting case $D=2$, the deformations vary as $\delta = G$ and $\delta_c = (K\phi/2)^2 a/G$. Therefore for plastic deformation to take place the condition of spot area is $a < G^2/(K\phi/2)^2$. For a smooth magnetic thin film rigid disk (Majumdar and Bhushan, 1990) typical values of the parameters are $D=1.38$, $G=10^{-16}$ m and $\phi=0.05$. The critical spot area for inception of plastic deformation is $a_c = 10^{-27}$ m² which is practically zero. Therefore all contact spots can be assumed to be in elastic contact at moderate loads.

The question now remains as to how do large spots become elastic when they must have been small plastic spots in their history of deformation. The possible explanation for this is graphically shown as follows in Fig. 6. When the contact plane is at position (a) then the two spots 1 and 2 are in plastic contact due to their small radius of curvature. However when the contact plane comes to position (b) the two spots merge to form a large spot 3 for which the radius of curvature is large enough, thereby relieving the stress for it to be in elastic contact. Of course there will be some work hardening involved, but that is a separate issue which is neglected in this model and will be considered in later studies. For the case of the untextured disk with $a_c = 10^{-27}$ m², the contact spots are always in elastic contact and therefore the change from plastic to elastic regimes is never encountered.

It is often found that the hardness of a surface is much higher than the bulk value and that the hardness is a function of the depth from the surface. This suggests that small contact spots, which are formed by small deformations of the surface, have larger hardness whereas larger spots corresponding to a larger depth from the surface have smaller hardness. Therefore, it can be concluded that the hardness is a function of the contact spot area. If this functional dependence is known, it can easily be incorporated in equation (8) for the critical deformation and in the load calculations, discussed now.

Surface Load and Separation. Consider a contact spot of area a . From the geometry of the asperity before deformation and the typical values of G (Majumdar and Tien, 1990) it can be assumed that $R \gg \delta$ from which follows the relation

$$a = \pi R \delta \quad (11)$$

If $a > a_c$ then the spot is in elastic deformation and therefore the elastic load P_e on such a spot is given by the Hertzian theory to be (Johnson, 1985)

$$P_e(\delta) = \frac{4}{3} ER^{1/2} \delta^{3/2} \quad (12)$$

Substituting for δ from equation (11) in equation (12)

$$P_e(a) = \frac{4Ea^{3/2}}{3\pi^{3/2}R} \quad (13)$$

In the G-W model the radius of all the asperities is assumed to be constant and equal to the mean radius of curvature. Equation (13) is then used in conjunction with the probability distribution of asperity heights to obtain the load-area relation. However equation (7) shows that R is scale-dependent and therefore cannot be assumed to be a constant. In this model, the relation for R in equation (7) is substituted in equation (13) to get the final form

$$P_e(a) = \frac{4\sqrt{\pi}EG^{D-1}a^{(3-D)/2}}{3} \quad (14)$$

One should note that the area-load relation for a hemisphere of radius R is $a \sim P^{2/3}$ according to the Hertzian theory. However, if the radius of curvature changes with the area of contact as in equation (7), the area-load relation follows equation (14), which is $a \sim P^{2/(3-D)}$. In the limiting cases of $D=1$ and $D=2$, the power-law behaviors are $a \sim P$ and $a \sim P^2$, respectively. This is in contrast with Archard's (1957) model of small hemispheres resting on bigger hemispheres, which rest on even bigger hemispheres, that yielded an exponent of load to be slightly less than one.

If the contact spot is in plastic deformation, that is if it satisfies $a < a_c$, then the load on the spot is given as

$$P_p(a) = K\sigma_y a \quad (15)$$

where σ_y is the yield strength of the material. It is interesting to observe that under elastic deformation of a surface of $D=1$, the linear area-load relation is indistinguishable from that under plastic deformation.

The loads required to make a single contact spot of area a has been determined in equation (14) and (15) for the regimes of elastic and plastic deformation, respectively. The total load on the surface can now be evaluated by using the size-distribution $n(a)$ of contact spots. If the largest contact spot is larger than the critical area, that is $a_l > a_c$, then the total load is given by

$$P = \frac{4\sqrt{\pi}EG^{D-1}}{3} \int_{a_c}^{a_l} n(a)a^{(3-D)/2} da + K\sigma_y \int_{a_s}^{a_c} n(a)ada \quad (16)$$

The first term in the above expression represents the elastic load and the second term, the plastic load. When $a_l < a_c$ then all the contact spots are in plastic contact and therefore the total load can be written simply as

$$P = K\sigma_y \int_{a_s}^{a_l} n(a)ada = K\sigma_y A_r \quad (17)$$

Note that if the functional dependence of the hardness $H = K\sigma_y$, on the contact spot area is known, it can easily be incorporated in the integrals of equations (16) and (17).

If the surface heights $z(x,y)$ follows a normal probability distribution, the separation d between the mean planes of the two surfaces can be obtained by the relation

$$\frac{A_r}{A_a} = \frac{1}{\sqrt{2\pi}} \int_{d/\sigma}^{\infty} \exp(-x^2/2) dx = \frac{1}{2} \operatorname{erfc}\left(\frac{d}{\sqrt{2}\sigma}\right) \quad (18)$$

where $\operatorname{erfc}()$ is the complimentary error function. Therefore, the separation and the load are related through the area-load relations of equations (16) or (17) and the area-separation relation of equation (18).

The main difference between the fractal model described above and the G-W model is that the fractal model considers the radius of curvature of a contact spot to be a function of the area of the spot. This has serious implications on the

elastic-plastic regime of contact spots. Since the G-W model assumes the same radius of curvature for all asperities, it is natural to find that asperities with higher deformation or larger contact area to be in plastic contact. However, it fails to account for the fact that the radius of curvature is a function of the asperity size. The fractal model shows that smaller asperities to be in plastic contact and larger ones to be in elastic contact.

To obtain results from the fractal model, the size-distribution of contact spots $n(a)$ is needed for solving the integral in equation (16). The following section discusses the size-distribution of a fractal surface.

4 Size-Distribution of Contact Spots

In the study of geomorphology of the earth, Mandelbrot (1975) found that the cumulative size-distribution of islands on the earth's surface follows the power-law $N(A > a) \sim a^{-D/2}$, where N is the total number of islands with area larger than a and D is the fractal dimension of the coastline of the islands. The coastline is a self-similar curve for which the dimension D is related to the surface dimension D_s by the relation $D = D_s - 1$. If the surface of, say, a machined metal or a thin-film magnetic disk, is enlarged enough, the hills and valleys appear similar to that found on the earth's surface. The contact spots formed by a horizontal cut of such a surface by a contact plane would be analogous to the islands formed by cutting the earth's surface by that of the ocean. Therefore, it can be expected that the contact spots would also follow a power law relation of the form

$$N(A > a) = \left(\frac{a_l}{a}\right)^{D/2} \quad (19)$$

where the distribution is normalized by the area of the largest contact spot a_l . One should note that the fractal contact model has been developed on the basis of the self-affinity dimension obtained from equation (2) which is also related to the surface dimension as $D = D_s - 1$. Since both the similarity dimension for the coastline and the affinity dimension of the profile are related in the same way to the surface dimension, the profile dimension can be used in equation (19). However, the number-area rule is proved only for the coastline dimension (Mandelbrot, 1982) and not for the self-affinity dimension of the surface profile. The relation between the spectral exponent in equation (2) and the area exponent in equation (19) needs to be investigated.

Note that in the distribution of equation (19) the number of the largest spot is unity whereas the number of spots of area $a \rightarrow 0$ would tend to infinity. Since the power spectrum in Fig. 4 shows that a surface can be fractal even at nanoscales, the assumption of $a_s \rightarrow 0$ is valid. Most existing contact theories commonly use the parameters asperity density η per unit area and the mean diameter of contact spots in their models. Since the number of spots tend to infinity, the values of η and the mean diameter would depend on the finite number of asperities and contact spots that are detectable by an instrument. Therefore both η and the mean spot diameter are scale-dependent parameters as has been analytically shown in Majumdar and Bhushan (1990). The fractal model does not involve these parameters and allows the smallest spot area to tend to zero with their number tending to infinity.

It was experimentally observed that the size-distribution of contact spots for magnetic tape and disks follows a power-law behavior of equation (19) (Majumdar and Bhushan, 1990). The size-distribution had two regimes of power laws - for larger spots the exponent suggested a dimension of 2 whereas for smaller spots the dimension was close to 1.15. This followed the spectrum of the surface which yielded a dimension of 1 for high frequencies or small roughness scales and 1.72 for low

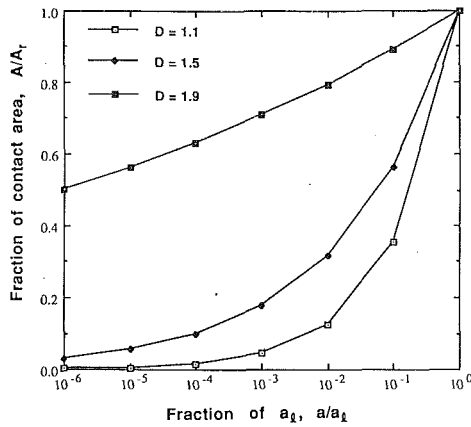


Fig. 7(a)

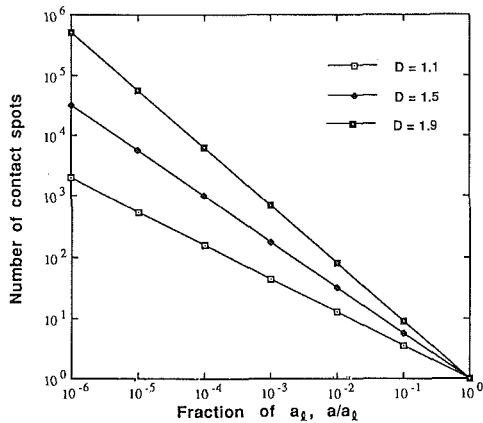


Fig. 7(b)

Fig. 7 The effect of fractal dimension of a surface profile on (a) the contribution all of contact spots of area less than a to the real area of contact (b) the number of spots of area larger than a

frequencies. When the load was increased from 69kPa to 1.38MPa, it was noted that the area of the largest spot increased in size whereas the distribution remained the same. From the cumulative distribution of equation (19), the distribution for $n(a)$ can be obtained by differentiation to get

$$n(a) = \frac{D}{2} \frac{a_i^{D/2}}{a^{D/2+1}} \quad (20)$$

Therefore the real area of contact A_r is given as

$$A_r = \int_0^{a_i} n(a) da = \frac{D}{2-D} a_i \quad (21)$$

for $D < 2$. For the case $a_i > a_c$, the portion of the real area of contact in elastic and plastic deformation can be evaluated as

$$A_{re} = \int_{a_c}^{a_i} n(a) da = \frac{D}{2-D} (a_i - a_c^{D/2} a_c^{(2-D)/2}) \quad (22a)$$

$$A_{rp} = \int_0^{a_c} n(a) da = \frac{D}{2-D} a_i^{D/2} a_c^{(2-D)/2} \quad (22b)$$

These analytical results in conjunction with experiments (Majumdar and Bhushan, 1990) suggest that the contact process can be described as follows—when the load is increased (i) the contact plane moves down from position (a) to (b), as shown in Fig. 6 (ii) small plastic spots merge to form larger elastic spots (iii) the area of the largest spot increases linearly with the total real area of contact (iv) new contact spots are

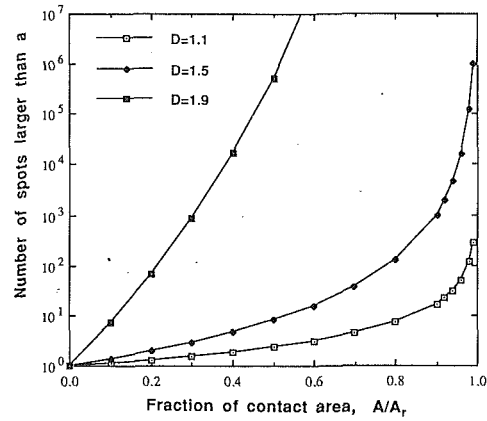


Fig. 8 Number of contact spots of area larger than a versus the fraction of the real area of contact contributed by the area of these spots larger than a

formed in such a way that the size-distribution remains the same.

Consider f to be that fraction of the real area of contact which is comprised of contact spots of area larger than a_f but smaller than a_i . The actual area A_f contributed by these spots can be evaluated from the size-distribution as

$$A_f = f A_r = \int_{a_f}^{a_i} n(a) da = \frac{D}{2-D} a_i \left(1 - \left(\frac{a_f}{a_i} \right)^{1-D/2} \right) \quad (23)$$

Comparison with equation (21) shows that the fraction f is equal to

$$f = 1 - \left(\frac{a_f}{a_i} \right)^{1-D/2} \quad (24)$$

Using equation (19) for the cumulative size-distribution, it follows that

$$N = \left(\frac{1}{1-f} \right)^{D/(2-D)} \quad (25)$$

Equation (25) suggests that for a given fraction of the real area of contact, the number of contact spots contributing to that fraction remains the same. The number is independent of the real area of contact and therefore the load. However, the sizes of the contact spots increase with load.

The contribution of a specific range of spot size to the total area of contact is also of importance in deciding the significance of the small or the big contact spots. Such a description is provided in Figs. 7(a, b) which show that although the number of contact spots increases for decreasing spot area, their contribution to the real area of contact may or may not become negligible. For $D = 1.5$, contacts spots of area between $0.1a_i$ and a_i contribute about 45 percent of the real area of contact. However, for $D = 1.9$ only 10 percent of the contact area is contributed by spots between $0.1a_i$ and a_i . For $D = 1.1$ spots between $0.1a_i$ and a_i contribute about 70 percent of total real contact area. So the contribution of a range of contact spots sizes to the total contact area depends on the fractal dimension of the surface. The number of spots of sizes larger than a , is shown in Fig. 7(b). The results show that the number can vary significantly depending on the dimension of the surface. Figure 8 shows the total number of contact spots larger than a versus the fraction of the real area of contact occupied by these spots. Note that for a particular fraction of the contact area, there is a drastic difference in the number of contact spots for different values of the dimension. Also, for a given number of contact spots larger than an area, the fraction of the real contact area covered by these spots is larger for

Table 1 Profile power spectrum, critical contact spot, size-distribution of spots, real area of contact, fraction of elastic contact area and total load-predictions of fractal contact model

Surface profile power spectrum	$S(\omega) = \frac{G^{2(D-1)}}{2 \ln \gamma} \frac{1}{\omega^{(5-2D)}}$
Plastic contact for spot $a < a_c$	$a_c = \frac{G^2}{(\pi K \phi / 2)^{2/(D-1)}}$
Number of spots larger than a	$N(A > a) = \left(\frac{a_l}{a} \right)^{D/2}$
Real area of contact	$A_r = \frac{D}{2-D} a_l$
Fraction of elastic contact area	$A_{re}/A_r = 1 - \left(\frac{D a_c}{(2-D) A_r} \right)^{(2-D)/2}$
Total elastic-plastic load ($a_l > a_c$)	$\frac{P}{EA_a} \sim G^{D-1} A_r^{D/2} \left[\left(\frac{(2-D) A_r}{D} \right)^{(3-2D)/2} - a_c^{(3-2D)/2} \right] + K \phi A_r^{D/2} a_c^{(2-D)/2}$ for $D \neq 1.5$ $\frac{P}{EA_a} \sim G^{1/2} A_r^{3/4} \ln(A_r/a_c) + K \phi A_r^{3/4} a_c^{1/4}$ for $D = 1.5$
Total plastic load ($a_l < a_c$)	$\frac{P}{EA_a} \sim K \phi A_r$

smaller dimension. This is in accordance with Fig. 7(a) which shows that for smaller dimension a large portion of the contact area is contributed by contact spots with $0.1a_l$ and a_l .

5 Results and Discussion

The size distribution of equation (20) can be used in equation (16) to obtain the load-area relation for the case when $a_l > a_c$. The integrals in equation (18) can be evaluated in the range $1 < D < 2$, to give the total load in a non-dimensional form as (for $D \neq 1.5$)

$$P^* = \frac{4\sqrt{\pi}}{3} G^{*(D-1)} g_1(D) A_r^{*D/2} \left[\left(\frac{(2-D) A_r^*}{D} \right)^{(3-2D)/2} - a_c^{*(3-2D)/2} \right] + K \phi g_2(D) A_r^{*D/2} a_c^{*(2-D)/2} \quad (26)$$

where

$$g_1(D) = \frac{D}{(3-2D)} \left(\frac{2-D}{D} \right)^{D/2}; \quad g_2(D) = \left(\frac{D}{2-D} \right)^{(2-D)/2} \quad (27)$$

and

$$P^* = \frac{P}{A_a E}; \quad G^* = \frac{G}{\sqrt{A_a}}; \quad A_r^* = \frac{A_r}{A_a}; \quad a_c^* = \frac{a_c}{A_a} \\ = \frac{D^*2}{(K \phi / 2)^{2/(D-1)}} \quad (28)$$

Equation (26) shows that there are basically four non-dimensional groups- P^* for the load, A_r^* for the real area of contact, G^* for the amplitude of roughness and ϕ the material property. The apparent area A_a can be determined from the spectral characterization parameter ω_l by the relation $A_a = \omega_l^{-2}$. The load P^* can now be evaluated as a function of A_r^* for different values of the material property ϕ , fractal dimension D and roughness parameter G^* .

For $D = 1.5$, the load-area relation in non-dimensional form is given as

$$P^* = \sqrt{\pi} G^{*1/2} \left(\frac{A_r^*}{3} \right)^{3/4} \ln \left[\frac{A_r^*}{3 a_c^*} \right] + \frac{3 K \phi}{4} \left(\frac{A_r^*}{3} \right)^{3/4} a_c^{*1/4} \quad (29)$$

If the largest contact spot a_l is less than the critical area for plastic deformation, $a_l < a_c$, then only plastic deformation will take place in which case the load is given by

$$P^* = K \phi A_r^* \quad (30)$$

Such a plastic loading neglects the interaction between asperities. Pullen and Williamson (1972) considered such interactions and found the load to vary as $P^* = \phi A_r^*/(1 - A_r^*)$. The interactions can be incorporated in the present model.

The first term in equation (26) represents the total elastic load whereas the second term, the plastic load. In the limiting case for $a_c^* \rightarrow 0$, as was observed for the magnetic thin-film rigid disk, the fractal model shows that the load-area relation reduces to $P^* \sim A_r^{*(3-D)/2}$. This power-behavior explains the origins of the exponent n which is discussed in the introduction.

The main results of the fractal model are listed in Table 1. The predictions for the real area of contact and the fraction of elastic contact area are graphically shown in Figs. 9(a), 10(a), and 11(a) for variation of one parameter keeping the others constant. Figures 9(b), 10(b), and 11(b) correspondingly show the fraction of the real area of contact in elastic deformation. For $D = 1.5$ and $\phi = 0.01$ in Fig. 9(a), it is evident that as G^* is increased, the load required to produce a particular real area of contact increases. This results from the fact that an increase in G^* implies an increase in asperity deformation δ , which therefore requires a higher load. Figure 9(b) shows that as G^* is increased the percentage of elastic contact decreases. This is caused due to a decrease in the radius of asperity curvature as evident in equation (7). Also, when the real area of contact is increased the fraction of elastic contact area increases. This

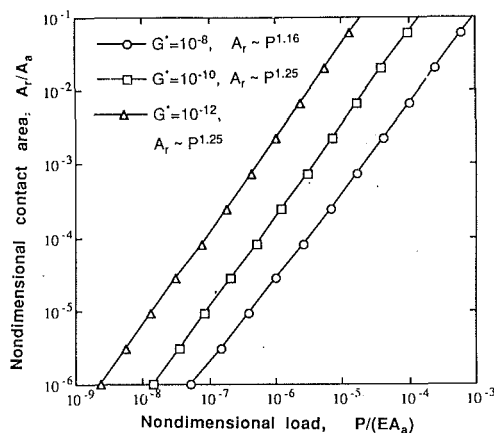


Fig. 9(a)

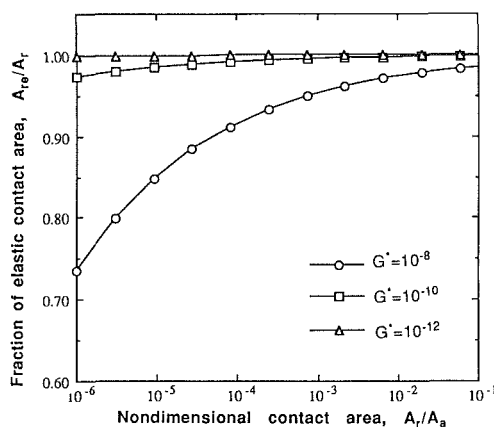


Fig. 9(b)

Fig. 9 Predictions of the fractal contact model for (a) the real area of contact and (b) fraction of real contact area in elastic deformation; for $D = 1.5$, $\phi = 0.01$ and different values of G^*

results from merging of small contact spots in plastic deformation to form larger spots in elastic deformation.

The effect of fractal dimension on the real area of contact and elastic contact area is interesting indeed and are shown in Fig. 10 for $G^* = 10^{-10}$ and $\phi = 0.01$. Consider the range of load $10^{-6} < P^* < 10^{-3}$ in Fig. 10(a). It is observed that as the value of dimension is increased, the real area of contact first increases till $D = 1.5$ and then decreases for $D = 1.9$ for a given load. A similar reversal behavior is observed for the fraction of elastic contact area in Fig. 10(b). The question now remains as to why is there a reversal behavior in the real contact area and the elastic contact area. As the dimension is increased from 1.1 to 1.9, the value of a_c^* decreases from 10^{16} to 10^{-16} , respectively, for the above values of G^* and ϕ . Therefore, for $D = 1.1$, all the contact spots are smaller than the critical contact area and so in plastic contact. This is evident in Fig. 10(b) which shows the fraction of elastic contact area to be zero. As D is increased, the value of a_c^* decreases which results in a large number of elastic contact spots. Therefore the fraction of elastic contact area for $D = 1.5$ is nearly unity. As the value of D is further increased, the critical area decreases further and one may expect that the fraction of elastic contact should increase. However, note that as D is increased, the contribution of smaller contact spots to the total area is significant and is evident in Fig. 7(a). This is due to a significant increase in the number of small contact spots as shown in Fig. 7(b). The increase in the value of fractal dimension leads to two effects—the critical contact area decreases and the number of contact spots below the critical size increases. Due to these competing effects, the fraction of elastic contact area decreases as the

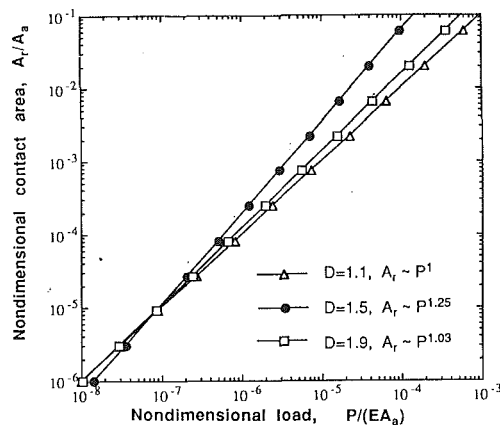


Fig. 10(a)

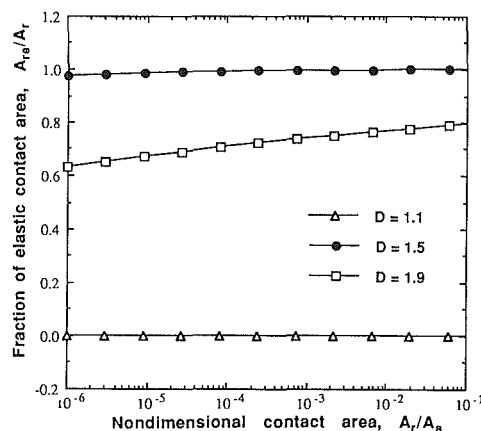


Fig. 10(b)

Fig. 10 Predictions of the fractal contact model for (a) the real area of contact and (b) fraction of real contact area in elastic deformation; for $G^* = 10^{-10}$, $\phi = 0.01$ and different values of D

dimension is increased to 1.9. This results in the reversal behavior evident in Figs. 10(a, b).

The effect of the material property ϕ is shown in Fig. 11, for $D = 1.5$ and $G^* = 10^{-10}$. It is observed that as ϕ is decreased there is a distinct change in the behaviors of both real contact area and elastic contact area. As ϕ is reduced, the value of a_c^* increases which results in larger degree of plastic contact area as evident in Fig. 11(b). Note that for $\phi = 10^{-4}$, the fraction of elastic contact area is zero for $A_r^* < 2 \times 10^{-5}$ and increases asymptotically to unity with larger A_r^* . This has a significant effect on the area-load relation in Fig. 11(a), which goes from purely plastic deformation at low loads to a combination of elastic-plastic deformation of contact spots at higher loads.

Comparison With Experiments. Figure 12(a) shows the experimental results for two Pyrex glass surfaces compressed against each other (Yamada et al., 1978). It can be observed that the area-load relation is $A_r^* \sim P^{1.25}$. The exponent corresponds to a fractal dimension of the surface profile to be 1.4 and of the surface to be 2.4. Note that for $P^* > 2 \times 10^{-3}$ the experimental results start to deviate from the predictions. It is speculated that this may be due to interactions between neighboring asperities in contact which has been neglected in the fractal model. This speculation is based on the fact that such an area-load behavior is similar to that predicted by the Pullen-Williamson (1972) model which assumes plastic deformation of interacting asperities under heavy loads. Figure 12(b) shows a comparison between experimental data obtained for magnetic tape A (Bhushan, 1985), the G-W model and the fractal model. The fractal model is in excellent agreement with

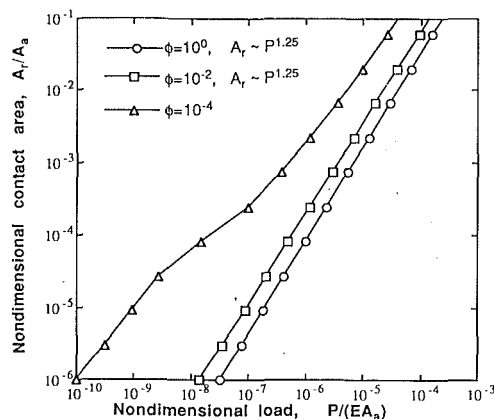


Fig. 11(a)

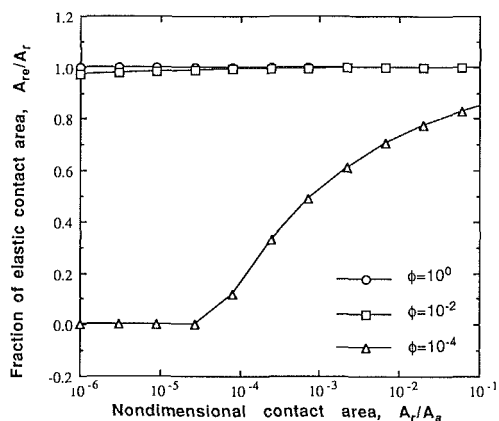


Fig. 11(b)

Fig. 11 Predictions of the fractal contact model for (a) the real area of contact and (b) fraction of real contact area in elastic deformation; for $G^* = 10^{-10}$, $D = 1.5$ and different values of ϕ

the data for $P^* < 2 \times 10^{-4}$, for values of $D = 1.49$, $G^* = 10^{-10}$ and $\phi = 0.05$. However, at higher loads, the interactions between the asperities result in a behavior similar to that observed in Fig. 12(a) where an increase in real area of contact is less than proportional to an increase of load. The G-W model predicts a linear load-area relationship which is not observed experimentally. Figure 12(c) shows the comparison between the predictions of the G-W model, the fractal model and experiments on contact between a magnetic thin-film rigid disk and a plano-convex lens (Bhushan and Dugger, 1990). The predictions of the fractal model are evaluated for experimentally observed values of $D = 1.38$ (Majumdar and Bhushan, 1990) and $\phi = 0.05$ (Bhushan and Doerner, 1989) for disk type C, and an assumed value for $G^* = 10^{-10}$ for curve fitting. It can be seen that firstly, the trends of the fractal model are in better agreement than that of the G-W model. In addition, the G-W model predicts a linear load-area relation which is not observed experimentally. However, one should note that having four data points in Fig. 12(c) may not provide enough evidence for the validity of the fractal model. In that respect, Fig. 12(a) provides a more conclusive agreement between theory and experiments.

6 Conclusions

This paper experimentally shows that a variety of surfaces are fractal in structure within millimeter scales and nanoscales. Based on such observations it is concluded that the statistical parameters such as the r.m.s. height, slope and curvature, which are conventionally used to characterize surface roughness, are scale-dependent. Since these parameters

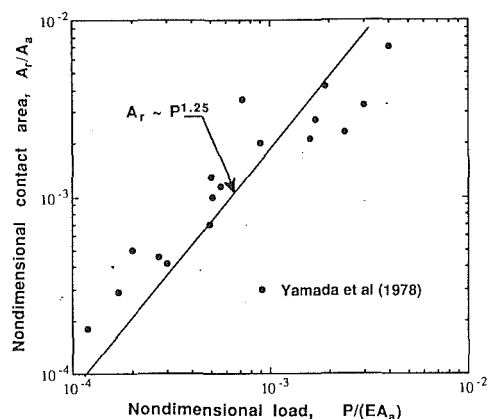


Fig. 12(a)

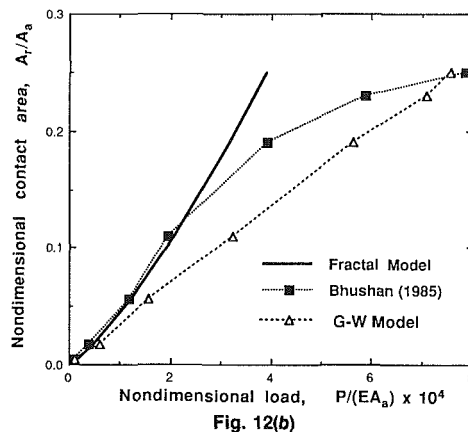


Fig. 12(b)

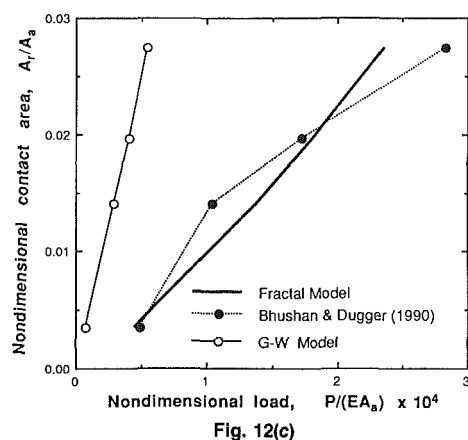


Fig. 12(c)

Fig. 12 Comparison of the fractal model and the G-W model with experiments

are extensively used in existing models of contact between rough surfaces, it is concluded that these models do not provide unique predictions. Surface roughness can be characterized by fractal geometry to yield *scale-independent* fractal parameters. Based on the fractal characterization, a new model for contact between two isotropic rough surfaces is developed. The model predicts that all contact spots of area less than a critical area, are in plastic contact. This result arises from the fact that smaller asperities have smaller radii of curvature and are therefore more likely to undergo plastic deformation. When the load is increased, these small plastic spots join to form elastic contact spots. This is in contrast with the classical Greenwood-Williamson model which assumes that all asperities to be of the same radius and predicts smaller contact spots to be in elastic deformation and larger spots in plastic deformation.

The fractal model uses a result from the geomorphology of the earth for the power-law size-distribution of contact spots. Using this distribution, the load-area relation for elastic contact is predicted to be $P \sim A_r^{(3-D)/2}$ where D is the fractal dimension of a surface profile and varies between 1 and 2. This result explains the origins of the area exponent which has been the focus of a number of studies in contacting surfaces and friction. The size-distribution of contact spots suggests that for a certain fraction of the real area of contact, the number of contact spots contributing to that fraction remains independent of the real area of contact and the load. However, the sizes of these contact spot increases with load. The model also shows that variations in the value of the fractal dimension and the material properties can have considerable effect on the load-area relations and the fraction of contact spot area in elastic deformation.

The predictions of the model are in reasonable agreement with experiments for low values of load. For higher loads, it is speculated that interactions between contacting asperities, which have been neglected in this model, play a significant role. A future study will generalize this fractal model by considering asperity interactions and anisotropic surfaces. In addition, consideration should be given to other effects such as work hardening due to elastic-plastic transition, variation of hardness with depth from a surface and to frictional forces between deforming asperities.

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